Junghyun Min

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SKILLS & SERVICE

Software Engineering: Unix, databases, Flask, FastAPI, asynchrony, Google Cloud, Docker, Portainer, Git, Jenkins.
Natural Language Processing: PyTorch, TensorFlow, transformers, TorchServe, scikit-learn, ML baselines
Large Language Models: OpenAI, LangChain, RAG, prompt engineering, vLLM, quantization, distributed training
Languages: Python, Java, C++, JavaScript, R, Wolfram; English/Korean (Native), German/Mandarin (Working)
Service: Reviewer for *ACL, Language Resources and Evaluation, Machine Learning Engineering; NACLO Site Host

EDUCATION

Georgetown University — Washington, DC
Ph.D. Computer Science and Linguistics. Advisor: Ethan Wilcox. — Exp. 2029

Johns Hopkins University — Baltimore, MD
M.A. Cognitive Science. Advisor: Tal Linzen. — Dec 2020

B.S. Physics, B.A. Mathematics. General Honors, early graduation. — Dec 2017

WORK EXPERIENCE

Nokia Bell Labs — Murray Hill, NJ
Machine Learning and AI Intern — Jun 2026 – Aug 2026
- Prototype and implement an agentic “autoresearch” pipeline for telecom ticket troubleshooting system improvement

University of Toronto — Toronto, ON
Computer Science Research Intern — May 2025 – Aug 2025
- Pre-trained LMs from scratch and via transfer; compiled 7-task Cantonese NLU benchmark and established baselines
- Performed large-scale verification of a novel training objective across architecture, language, and resource settings

NCSOFT — Seongnam, Korea
Machine Learning Engineer (Natural Language Processing) at Language AI Lab — Jan 2021 – Apr 2024
- Implemented and trained tokenizers for the Language Model Task Force, precursor to the 1.3-13B VARCO LLMs
- Deployed efficient, scalable Stanza-like cloud REST API, parsing 10k requests per sec on 4GB VRAM at 96% acc.
- Served machine learning models for information retrieval via Docker and TorchServe, processing 30+ sent per sec.
- Designed algorithms in deep learning models to improve information extraction performance (+30%p) and stability
- Built and optimized agentic pipelines to automatically parse current news to extract and rank 1,000 events per day

SELECTED TECHNICAL PROJECTS

Lead engineer, Agentic document automation platform genDOC
- Built RAG-based agentic document automation software with LangChain and FastAPI on OpenAI API using Python
- Implemented orchestration framework, routing pipeline, agent-wise output verification logic and self-correction loops
- Optimized RAG pipeline to achieve 90+% end-to-end pass rate (pass@5) and end-to-end latency under 60 seconds

Lead software engineer, ai.ly
- Fine-tuned and cloud deployed a GPT-based AI lyricist that generates lyrics aligned to user preferences on PyTorch
- Accumulated 50k+ visits over 3 months of service, cross-functional collaboration resulting in a hip-hop sample

Co-lead, Large language models and legal interpretation
- Implemented LLM legal interp. judgment extraction tool with vLLM. Oral presentation, NLLPW at EMNLP 2025
- Showed even large (70B) and commercial LLMs are sensitive to surface form, lack alignment to human judgment

Lead, Multimodal text-prosody model for syntactic information content estimation
- Developed and fine-tuned 3-modal transformer architecture for syntax prediction with multimodal text and audio input
- Measured syntax information content in text (-74%), duration (-3%), pause (-1.7%) in naturalistic and planned speech

SELECTED PUBLICATIONS

- **Junghyun Min**, Minho Lee, Woochul Lee, Yeonsoo Lee. RepL4NLP at NAACL 2025. Punctuation Restoration Improves Structure Understanding without Supervision. Tech blog (Korean). PyTorch implementation on Docker.
- Abhishek Purushothama, **Junghyun Min**, Brandon Waldon, Nathan Schneider. FAccT 2026. Prompting from the bench: Large-scale pretraining is not sufficient to prepare LLMs for ordinary meaning analysis. vLLM and OpenAI API.
- **Junghyun Min**, R. Thomas McCoy, Dipanjan Das, Emily Pitler, and Tal Linzen. ACL 2020. Syntactic data augmentation increases robustness to inference heuristics. Collaboration with Google Research using TensorFlow.